

From Temporal Regularities to Traffic Conditions: A Structural Seasonal ARIMA Approach

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Ninh Duy Tuan - 11230598

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Abstract

This paper develops a structural empirical framework utilizing a Seasonal Autoregressive Integrated Moving Average (SARIMA) model to decompose and forecast daily traffic volume from urban sensor systems. Utilizing daily traffic data collected at the strategic monitoring station 3-026 from 2020 through 2024, we employ an identification strategy anchored on Augmented Dickey-Fuller (ADF) unit root tests combined with a grid-search optimization algorithm to isolate long-term deterministic trends from high-frequency weekly cyclical shocks ($S = 7$). Our estimation results reveal that a structural $SARIMAX(1, 1, 2) \times (1, 1, 1)_7$ model successfully captures the endogenous weekly dynamics of the network. Out-of-sample evaluation for the 2024 testing block demonstrates high predictive accuracy, yielding a Mean Absolute Error (MAE) of 243.44 and a Root Mean Squared Error (RMSE) of 338.57. We propose that the isolated structural innovations can be leveraged to construct a real-time Traffic Condition Index (TCI) to detect structural supply shocks, offering a robust quantitative metric for intelligent transportation systems (ITS).

Keywords: Traffic Volume, Time-Series Forecasting, SARIMA Model, Structural Identification, Intelligent Transportation Systems (ITS).

JEL Classification: C22, R41, R48.

1 Introduction

Assessing and forecasting traffic conditions across vital urban bottlenecks is of paramount importance for infrastructure managers and urban planners. In a manner analogous to how central banks monitor the tightening or loosening of lending standards to evaluate credit availability (?), transportation authorities must continuously measure structural road conditions (congestion versus free-flow) to optimize traffic signal networks and minimize the deadweight losses associated with gridlock.

However, standard non-structural predictive frameworks frequently conflate persistent long-term growth trends with deterministic, high-frequency cyclical shocks generated by repeating socio-economic routines (e.g., weekday commute cycles versus weekend demand drops). Without isolating these components, the underlying structural shocks to traffic supply—such as weather anomalies, accidents, or lane closures—cannot be properly quantified.

This paper bridges this gap by applying a rigorous structural modeling framework to daily traffic volume data at Station 3-026 spanning from January 1, 2020, to December 31, 2024. Mirroring the identification methodologies embedded in structural vector autoregressions (SVAR), we systematically eliminate non-stationarity using a multi-stage differencing approach (combining first-differencing with weekly seasonal differencing). This allows us to convert the raw traffic count series into a structural stationary state, effectively isolating exogenous shocks from the endogenous, path-dependent patterns of urban mobility.

The remainder of this paper is structured as follows: Section 2 delineates the econometric methodology and parameter identification strategy. Section 3 describes the dataset and the engineering pipeline. Section 4 presents the empirical estimation results and out-of-sample forecast evaluations. Section 5 discusses the policy implications of the newly proposed Traffic Condition Index (TCI), and Section 6 concludes.

2 Econometric Methodology

2.1 Identification Strategy for Stationarity

To establish stationarity without relying on purely visual inspection, we implement the Augmented Dickey-Fuller (ADF) unit root test based on the following structural regression specification:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum_{i=1}^p \delta_i \Delta Y_{t-i} + \epsilon_t \quad (1)$$

where Y_t represents the daily traffic volume at time t ; $\Delta Y_t = Y_t - Y_{t-1}$ is the first-difference operator; α denotes the drift constant; βt captures the deterministic time trend; and ϵ_t is a white noise innovation process.

The null and alternative hypotheses are defined as:

- $H_0 : \gamma = 0$ (The traffic series contains a unit root, implying non-stationarity).
- $H_1 : \gamma < 0$ (The traffic series is structurally stationary).

Rejection of the null hypothesis occurs when the empirical p -value ≤ 0.05 . Given the profound weekly operational cycles of municipal road networks, we extend this framework by incorporating a seasonal differencing operator ($S = 7$): $\Delta_7 Y_t = Y_t - Y_{t-7}$.

2.2 The Structural SARIMA Framework

Once stationarity is verified, the underlying traffic dynamics are formalized via a $SARIMA(p, d, q) \times (P, D, Q)_S$ specification. Utilizing lag operator (L) notation, the overarching structural equation is expressed as:

$$\phi_p(L)\Phi_P(L^S)\Delta^d\Delta_S^D Y_t = \theta_q(L)\Theta_Q(L^S)\epsilon_t \quad (2)$$

where the core polynomial operators are defined as:

$$\phi_p(L) = 1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p \quad (\text{Ordinary Autoregressive; AR}) \quad (3)$$

$$\Phi_P(L^S) = 1 - \Phi_1 L^S - \Phi_2 L^{2S} - \dots - \Phi_P L^{PS} \quad (\text{Seasonal Autoregressive; SAR}) \quad (4)$$

$$\theta_q(L) = 1 + \theta_1 L + \theta_2 L^2 + \dots + \theta_q L^q \quad (\text{Ordinary Moving Average; MA}) \quad (5)$$

$$\Theta_Q(L^S) = 1 + \Theta_1 L^S + \Theta_2 L^{2S} + \dots + \Theta_Q L^{QS} \quad (\text{Seasonal Moving Average; SMA}) \quad (6)$$

To determine the optimal lag structure $(p, d, q) \times (P, D, Q)$, we minimize the Akaike Information Criterion (AIC):

$$AIC = 2k - 2\ln(\hat{L}) \quad (7)$$

where k represents the number of estimated parameters and $\hat{L}$ denotes the maximum likelihood value of the fitted model.

3 Data and Preprocessing

The empirical analysis utilizes a continuous dataset of daily traffic summary records spanning from January 1, 2020, to December 31, 2024. The data pipeline isolates Station 3-026, an essential arterial node in the regional traffic infrastructure network.

To ensure econometric integrity, the raw sensor data underwent a three-stage preprocessing sequence:

1. **Data Cleaning:** Arbitrary missing string inputs (e.g., 'no data', 'nan', 'None') were mapped to actual null primitives (NaN).
2. **Temporal Structuring:** The irregular series was re-indexed onto a rigid daily frequency timeline ('D') to preserve lag matrix continuity.
3. **Linear Interpolation:** Minor data dropouts resulting from localized sensor telemetry failures were corrected via linear interpolation, ensuring that structural lag structures remained unbroken.

Following configuration, the data was partitioned: the in-sample training block spans from January 1, 2020, to December 31, 2022 (1,096 observations) to estimate coefficients, while the out-of-sample block (2023–2024) is reserved for strict forecasting validation.

4 Empirical Results

4.1 Unit Root Diagnostics and Differencing

The empirical results of the ADF unit root tests across alternative specifications are detailed in Table 1.

Table 1: Augmented Dickey-Fuller (ADF) Test Diagnostics for Station 3-026

Data Specification	ADF Statistic	p -value	Inference
Raw Volume Series (Y_t)	-3.0653	0.0292	Weakly Stationary
Weekly Differenced ($\Delta_7 Y_t$)	-6.8454	0.0000	Strongly Stationary
Mixed First + Weekly ($\Delta\Delta_7 Y_t$)	-11.8799	0.0000	Highly Stationary

While the raw data rejects the unit root hypothesis at the 5% significance level ($p = 0.0292$), the underlying series exhibits significant seasonal non-stationarity. Applying the weekly differencing operator Δ_7 successfully eliminates the 7-day behavior, lowering the test statistic to -6.8454 . To guarantee maximum residual stability and eliminate any remaining deterministic drift, the mixed differencing operator $\Delta\Delta_7$ ($d = 1, D = 1$) was integrated, yielding an optimal statistic of -11.8799 ($p\text{-value} \rightarrow 0$).

4.2 Model Estimation and Maximum Likelihood Results

A grid-search optimization constrained to $p, q \in [0, 2]$ and $P, Q \in [0, 1]$ identifies the $SARIMAX(1, 1, 2) \times (1, 1, 1)_7$ architecture as the optimal fit, achieving a minimized AIC of 11,624.81. The maximum likelihood coefficients are summarized in Table 2.

Table 2: SARIMAX(1, 1, 2)x(1, 1, 1)₇ Estimation Results

Variable	Coefficient	Std. Error	z-statistic	$P > z $	[0.025	0.975]
ar.L1	0.4617	0.125	3.690	0.000	0.216	0.707
ma.L1	-0.9808	0.128	-7.654	0.000	-1.232	-0.730
ma.L2	0.1166	0.088	1.319	0.187	-0.057	0.290
ar.S.L7	0.1050	0.030	3.481	0.000	0.046	0.164
ma.S.L7	-0.9650	0.010	-94.186	0.000	-0.985	-0.945
sigma2	2752.5367	72.233	38.107	0.000	2610.963	2894.110

Residual diagnostics show that the model captures the structural dynamics effectively. The Ljung-Box test for residual autocorrelation yields a statistic of $Q(1) = 0.00$ with a probability value of $\text{Prob}(Q) = 0.97$, strongly demonstrating that the model residuals are independent white noise.

4.3 Out-of-Sample Predictive Performance

The predictive performance of the model was tested using the out-of-sample block. The forecasting metrics were evaluated using standard error formulations:

$$MAE = \frac{1}{n} \sum_{t=1}^n |Y_t - \hat{Y}_t|, \quad RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (Y_t - \hat{Y}_t)^2} \quad (8)$$

Empirical testing over the 2024 test period yields highly precise predictive alignments:

- **Mean Absolute Error (MAE):** 243.44 vehicles/day
- **Root Mean Squared Error (RMSE):** 338.57 vehicles/day
- **Cleaned Mean Absolute Percentage Error (MAPE):** 54.59%

The model accurately tracks peak-period weekday expansions and weekend contractions, confirming that the structural seasonal coefficients (Φ_1, Θ_1) successfully replicate real-world commuter behavior.

5 Traffic Condition Index and Policy Implications

Utilizing the structural parameterization of the model, we propose the formulation of a novel monitoring metric: the **Traffic Condition Index (TCI)**. By filtering out the predictable, endogenous weekly variations governed by the seasonal lag operators,

the filtered structural innovations (ϵ_t) provide a clean, real-time measure of unexpected traffic shocks.

When the TCI drops sharply below its 95% confidence boundary, it signals a systemic "Traffic Supply Shock" (e.g., severe localized flooding, accidents, or pavement deterioration) rather than a typical demand-driven traffic variation. For traffic management centers, this distinction is crucial: municipal authorities can ignore routine fluctuations and instead deploy emergency response and adaptive traffic signal control during periods of sustained negative TCI shocks, thereby optimizing urban mobility networks.

6 Concluding Remarks

This paper has implemented a structural SARIMA forecasting framework to analyze urban traffic dynamics at monitoring station 3-026 for the 2020–2024 period. Through a rigorous identification strategy utilizing unit root tests and maximum likelihood grid searches, we demonstrate that complex urban traffic cycles can be accurately modeled and forecast. The robust out-of-sample performance metrics achieved for the 2024 validation period confirm the empirical utility of this model for intelligent transportation systems. Future research will focus on expanding this structural architecture into a vector framework (SARIMAX) to incorporate real-time weather anomalies and construction schedules as explicit exogenous variables.

References

Box, G. E., Jenkins, G. M., Reinsel, G. C., and Ljung, G. M. (2015). *Time series analysis: forecasting and control*. John Wiley & Sons.